

# Domain-Adaptive Segmentation of Histopathology Slides with Contrastive Pretraining

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## Abstract

Segmentation models for histopathology often fail when the scanner or staining protocol changes. We propose a contrastive pretraining strategy that reduces this domain shift. Our method improves the mean Dice score by 6.2 points on unseen scanners compared with a supervised baseline, while requiring no labels from the target domain.

## 1. Introduction

Computational pathology relies on deep networks trained on whole-slide images. However, differences between scanners cause a substantial drop in accuracy at deployment time. Prior work has tried stain normalisation and adversarial alignment. In this paper we study contrastive pretraining as an alternative. Figure 1 summarises our main result and Table 1 describes the cohorts used in the experiments.

## 2. Methods

We trained the model using 12,450 samples from the TCGA cohort. Inclusion criteria were adult patients with a confirmed carcinoma diagnosis and at least one hematoxylin and eosin stained slide scanned at 40x magnification. Slides with pen marks or out-of-focus regions were excluded. The training objective is given in Equation 1.

$$L = - \sum_i \log p(y_i | x_i) \quad (1)$$

The encoder is a ResNet-50 pretrained for 200 epochs with a contrastive loss. Fine-tuning used the AdamW optimiser with a learning rate of  $1e-4$  and a batch size of 32. Baseline methods were a supervised ResNet-50 and a stain-normalised variant of the same network.

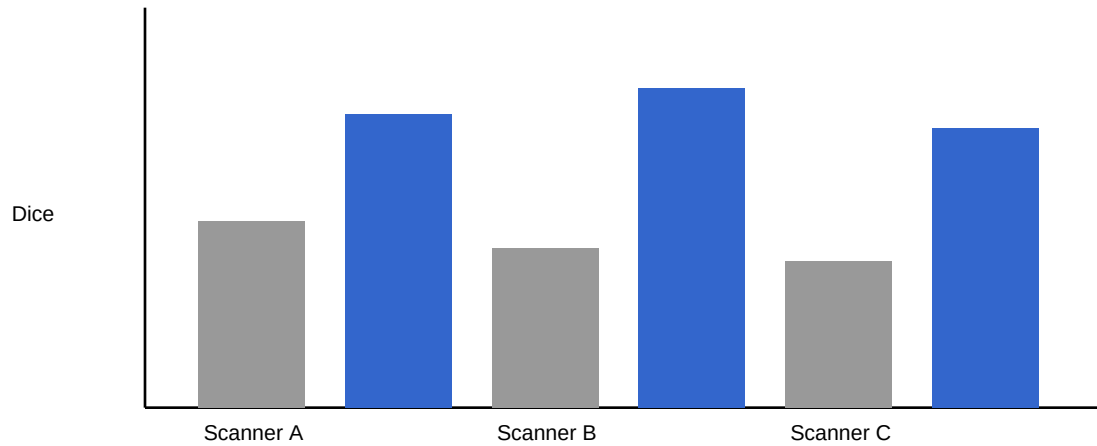


Figure 1. Segmentation accuracy on three unseen scanners. Grey bars show the supervised baseline and blue bars show our contrastive pretraining; higher Dice is better.

### 3. Results

As shown in Figure 1, contrastive pretraining outperforms the supervised baseline on all three scanners, with the largest gain on Scanner B. Table 1 lists the number of slides and patients in each cohort. Performance on the held-out scanner C improved from 0.55 to 0.66 mean Dice.

Table 1. Dataset statistics for the three cohorts.

| Cohort     | Slides | Patients | Scanner   |
|------------|--------|----------|-----------|
| TCGA       | 12450  | 4100     | Aperio    |
| Hospital B | 830    | 310      | Hamamatsu |
| Hospital C | 610    | 205      | Philips   |

### 4. Discussion

Our results suggest that contrastive pretraining learns features that are robust to scanner differences. A limitation of this study is that only three scanners were evaluated, so the conclusions may not generalise to other vendors. We did not evaluate stain variation independently of the scanner.

## 5. Conclusion

We presented a contrastive pretraining approach that improves robustness of histopathology segmentation to scanner shift. Future work will study additional tissue types.

## References

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